DEPARTMENT OF

MECHANICAL ENGINEERING

VISION OF THE DEPARTMENT

To develop into a Centre of Excellence in Education and interdisciplinary research with cutting edge technologies in the field of Mechanical Engineering, consistent with the contemporary and future societal needs of the country

MISSION OF THE DEPARTMENT

- To impart high quality education by using modern pedagogical tools so as to make the students technically competent in their chosen fields.
- To inculcate quality research by developing linkages with Industry and R & D organizations in India & abroad for developing technically competent and socially responsible engineers, managers and entrepreneurs.

B.TECH.
(MECHANICAL ENGINEERING)

B.TECH. (ME)

PROGRAM EDUCATIONAL OBJECTIVES

PEO-I: To prepare students for successful careers as mechanical engineers in organizations that meet the needs of Indian and global/multinational industrial/research establishments.

PEO-II: To provide a strong foundation in mathematical, scientific and engineering fundamentals in both domain and cross domain spheres, that enables students to visualize, analyze and solve mechanical engineering problems and be innovative and research oriented.

PEO-III: To train students with a wide spectrum of scientific and Mechanical engineering courses so that students could comprehend, analyze, design and create products and services, which promotes entrepreneurship culture in the campus that address real life problems, which are efficient and cost effective.

PEO-IV: To inculcate professional and ethical attitude in students, impart effective communication skills and ability to work in teams with multidisciplinary approach, be part of and interact with professional bodies so as to resolve engineering issues of social relevance.

PEO-V: To provide students with an academic environment that fosters excellence, leadership, yearning to pursue higher studies and passion for lifelong learning so as to have a successful professional career.

B.TECH. (ME)

PROGRAM OUTCOMES

PO-1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO-2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4).

PO-3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5).

PO-4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO-5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6).

PO-6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)

PO-7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9).

PO-8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO-9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO-10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO-11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8).

B.TECH. (ME)

PROGRAM SPECIFIC OUTCOMES

PSO-1: Apply expertise in design, manufacturing, materials and energy systems to develop innovative and sustainable solutions that drive industrial growth, ensure safety, promote societal well-being, and support green practices.

PSO-2: Create innovative research, develop sustainable products, formulate managerial and entrepreneurial strategies to implement emerging technologies for addressing industrial, societal, and global challenges.

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. I YEAR
MECHANICAL ENGINEERING

I SEMESTER **R25**

Course Code	Title of the Course	L	T	P/D	CH	C
25BS1MT101	Matrices and Calculus	3	1	0	4	4
25BS1PH101	Applied Physics	3	0	0	3	3
25HS1EN101	English for Skill Enhancement	3	0	0	3	3
25ES1ME101	Engineering Mechanics	3	0	0	3	3
25PC1ME101	Manufacturing Processes	3	0	0	3	3
25BS2PH101	Applied Physics Laboratory	0	0	2	2	1
25HS2EN101	English Language and Communication Skills Laboratory	0	0	2	2	1
25PC2ME101	Manufacturing Processes Laboratory	0	0	2	2	1
25ES2ME101	Engineering Workshop	0	0	2	2	1
25MN6HS101	Induction Programme	2	0	0	2	0
Total		**17**	**1**	**8**	**26**	**20**

II SEMESTER **R25**

Course Code	Title of the Course	L	T	P/D	CH	C
25BS1MT105	Ordinary Differential Equations and Vector Calculus	2	1	0	3	3
25BS1CH102	Engineering Chemistry	3	0	0	3	3
25ES1EE104	Elements of Electrical and Electronics Engineering	2	0	0	2	2
25ES1AM101	C Programming and Data Structures	2	0	0	2	2
25PC1ME102	Thermodynamics	3	0	0	3	3
25ES3ME102	Engineering Graphics	0	0	6	6	3
25BS2CH101	Engineering Chemistry Laboratory	0	0	2	2	1
25ES2AM101	C Programming and Data Structures Laboratory	0	0	4	4	2
25ES2EE104	Elements of Electrical and Electronics Engineering Laboratory	0	0	2	2	1
25MN6HS102	Environmental Science	2	0	0	2	0
Total		**14**	**1**	**14**	**29**	**20**

L – Lecture T – Tutorial P – Practical D – Drawing CH – Contact Hours/Week
C – Credits SE – Sessional Examination CA – Class Assessment ELA – Experiential Learning Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation LR – Lab Record
CP – Course Project PE – Practical Examination

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. I Semester

(25BS1MT101) MATRICES AND CALCULUS

TEACHING SCHEME		
L	T/P	C
3	1	4

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Mathematical Knowledge at Pre-University level

COURSE OBJECTIVES:
- To learn the concept of a rank of the matrix and applying this concept to check the consistency of a system of linear equations
- To learn the concept of eigen values and eigen vectors and to reduce the quadratic form to canonical form
- To know the geometrical approach to the mean value theorems and their application to the mathematical problems
- To find the maxima and minima of function of two and three variables
- To evaluate multiple integrals and their applications

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Compute the rank of a matrix and analyze the solution of a system of linear equations
CO-2: Calculate the Eigenvalues and Eigen vectors and Reduce the quadratic form to canonical form using orthogonal transformations
CO-3: Solve the applications on the mean value theorems
CO-4: Find the extreme values of functions of two and three variables with/ without constraints
CO-5: Evaluate the multiple integrals and apply the concept to find areas, volumes

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	2	2	1	1	1	1	-	2	1	2	2	1
CO-2	3	2	2	1	2	1	1	-	2	1	2	2	1
CO-3	3	2	2	1	1	1	1	-	2	1	2	2	1
CO-4	3	2	2	1	2	1	1	-	2	1	2	2	1
CO-5	3	2	2	1	2	1	1	-	2	1	2	2	1

UNIT-I:
Matrices: Rank of a matrix by Echelon form and Normal form – Inverse of Non-singular matrices by Gauss-Jordan method. System of linear equations: Solving system of Homogeneous and Non-Homogeneous equations, Gauss Seidel Iteration Methods.

UNIT-II:
Eigenvalues and Eigenvectors: Eigenvalues – Eigenvectors and their properties – Diagonalization of a matrix – Cayley-Hamilton Theorem (without proof) – finding inverse and power of a matrix by Cayley-Hamilton Theorem – Quadratic forms and

Nature of the Quadratic Forms – Reduction of Quadratic form to canonical forms by Orthogonal Transformation.

UNIT-III:

Single Variable Calculus: Limits and Continuous function and its properties. Mean value theorems: Rolle's theorem – Lagrange's Mean value theorem with their Geometrical Interpretation and applications – Cauchy's Mean value Theorem – Taylor's Series (All the theorems without proof).

Curve Tracing: Curve tracing in cartesian and polar coordinates

UNIT-IV:

Partial Differentiation and Its Applications: Definitions of Limit and continuity – Partial Differentiation: Euler's Theorem – Total derivative – Jacobian – Functional dependence & independence. Applications: Maxima and minima of functions of two variables and three variables using method of Lagrange multipliers.

UNIT-V:

Multiple Integrals: Evaluation of Double Integrals (Cartesian and polar coordinates) – change of order of integration (only Cartesian form) – Change of variables for double integrals (Cartesian to polar). Evaluation of Triple Integrals – Change of variables for triple integrals (Cartesian to Spherical and Cylindrical polar coordinates). Applications: Areas by double integrals and volumes by triple integrals.

TEXT BOOKS:

1. Higher Engineering Mathematics, B. S. Grewal, 36th Edition, Khanna Publishers, 2010
2. Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, 5th Editon, Narosa Publications, 2016

REFERENCES:

1. Advanced Engineering Mathematics, Erwin Kreyszig, 13th Edition, John Wiley & Sons, 2006
2. Thomas' Calculus, G. B. Thomas, 13th Edition, Pearson, 2014
3. A Text Book of Engineering Mathematics, N. P. Bali and Manish Goyal, Laxmi Publications, Reprint, 2008
4. Higher Engineering Mathematics, H. K. Dass and Er. Rajnish Verma, S. Chand

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. I Semester

(25BS1PH101) APPLIED PHYSICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: 10 + 2 Physics

COURSE OBJECTIVES:

- To explore the working and applications of lasers and fibre optics in modern technology
- To study crystal structures and defects in materials
- To learn nano materials and characterization techniques like XRD & SEM
- To understand the properties and applications of dielectric and magnetic materials
- To identify the importance of quantum mechanics and quantum computing

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Explain the principles of lasers and fibre optics and their applications in communication and sensing
CO-2: Understand different types of crystals and the importance of crystal defects
CO-3: Identify the role of nanomaterials in science and engineering applications
CO-4: Classify the dielectric and magnetic materials to illustrate their applications
CO-5: Realize the role of quantum mechanics in quantum computing

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	2	1	1	-	2	1	-	-	-	1	2	-
CO-2	3	3	-	-	-	-	1	-	-	-	1	1	-
CO-3	3	2	2	1	1	2	1	-	-	-	1	-	-
CO-4	3	3	1	1	1	2	1	-	-	-	1	1	-
CO-5	3	3	1	1	1	2	1	-	-	-	1	-	-

UNIT-I:

Laser and Fibre Optics: Introduction to laser, characteristics of laser, Einstein coefficients and their relations, metastable state, population inversion, pumping, lasing action, Ruby laser, He-Ne laser, Semiconductor laser, applications: Bar code scanner, LIDAR for autonomous vehicle.

Introduction to fibre optics, total internal reflection, construction of optical fibre, acceptance angle, numerical aperture, classification of optical fibres, losses in optical fibre, applications: optical fibre for communication system, sensor for structural health monitoring.

UNIT-II:

Crystallography & Defects: Introduction: Unit cell, space lattice, basis, lattice parameters; crystal structures, Bravais lattices, packing factor: SC, BCC, FCC; Miller indices, inter-planar distance; Introduction to defects, Point defects (Vacancies, Interstitial and Impurities) Line imperfections, Edge and Screw dislocation, Burger vector, Surface defects and volume defects (Qualitative Treatment).

UNIT-III:

Nano Materials & Characterization: Concept of Nanomaterials: surface to volume ratio and quantum confinement, Fabrication: Sol-Gel, Ball milling, Applications. **Characterization:** X -ray diffraction, Bragg's law, powder method, Calculation of average crystallite size using Debye Scherrer's formula, scanning electron microscopy (SEM): block diagram, working principle.

UNIT-IV:

Dielectric and Magnetic Materials: Introduction to dielectric materials, types of polarization (qualitative): electronics, ionic & orientation; ferroelectric, piezoelectric, pyroelectric materials and their applications: Ferroelectric Random-Access Memory (Fe-RAM), load cell and fire sensor.

Introduction to magnetic materials, classification of magnetic materials, hysteresis, Weiss domain theory of ferromagnetism, soft and hard magnetic materials, applications: magnetic hyperthermia for cancer treatment, magnets for EV, Giant Magneto Resistance (GMR) device.

UNIT-V:

Quantum Mechanics and Quantum Computing: Introduction, de-Broglie hypothesis, Heisenberg uncertainty principle, physical significance of wave function, postulates of quantum mechanics: operators in quantum mechanics, eigen values and eigen functions, expectation value; Schrödinger's time independent wave equation, particle in a 1D box.

Introduction, concept of quantum computer, advantages of quantum computing over classical computation. classical bits, Qubits, multiple Qubit system, quantum computing system for information processing, Dirac's Bra and Ket notation and their properties, Introduction to entanglement.

TEXT BOOKS:

1. Textbook of Engineering Physics, M. N. Avadhanulu, P. G. Kshirsagar & T. V. S. Arun Murthy, 11th Edition, S. Chand, 2019
2. Introduction to Solid State Physics, Charles Kittel, John Wiley & Sons
3. Introduction to Classical and Quantum Computing, Thomas G. Wong, Rooted Grove

REFERENCES:

1. Engineering Physics, B. K. Pandey and S. Chaturvedi, 2nd Edition, Cengage Learning, 2022
2. Energy Materials a Short Introduction to Functional Materials for Energy Conversion and Storage, A. S. Bandarenka, CRC Press, Taylor & Francis Group Energy Materials, Taylor & Francis Group, 1st Edition, 2022
3. Concepts of Modern Physics, Aurther Beiser, McGraw-Hill

4. Engineering Physics, P. K. Palanisamy, SciTech Publications
5. Quantum Computing, Jozef Gruska, McGraw-Hill

ONLINE RESOURCES:

1. https://shijuinpallotti.wordpress.com/wp-content/uploads/2019/07/optical-fiber-communications-principles-and-pr.pdf
2. https://www.geokniga.org/bookfiles/geokniga-crystallography_0.pdf
3. https://dpbck.ac.in/wp-content/uploads/2022/10/Introduction-to-Solid-State-PhysicsCharles-Kittel.pdf
4. https://www.thomaswong.net/introduction-to-classical-and-quantum-computing-1e4p.pdf

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. I Semester

(25HS1EN101) ENGLISH FOR SKILL ENHANCEMENT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Basic English

COURSE OBJECTIVES:
- To enhance vocabulary through word formation processes
- To read and comprehend different kinds of texts
- To write clear, concise paragraphs, essays, and letters to produce appropriate technical prose
- To improve coherence and cohesion in writing and speaking
- To recognize and practice the use of rhetorical elements necessary for the successful practice of scientific and technical communication

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Use vocabulary contextually and effectively in oral and written communication
CO-2: Demonstrate an understanding of the rules of functional Remedial Grammar and sentence structures
CO-3: Demonstrate improved competence in Standard Written English, including Remedial Grammar, sentence and paragraph structure, and coherence, and use this knowledge to accurately communicate technical information
CO-4: Apply principles of writing while developing paragraphs, essays, letters and technical prose
CO-5: Employ appropriate rhetorical patterns of discourse in scientific and technical communication

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	1	1	1	1	-	1	1	1	2	3	2	1	-
CO-2	2	2	2	2	2	2	2	1	3	3	2	-	-
CO-3	2	2	2	2	1	3	2	1	3	3	2	1	2
CO-4	1	1	1	1	1	2	2	1	2	3	2	-	-
CO-5	1	1	1	1	-	2	1	1	2	2	1	-	2

UNIT-I:
1. **Reading:** -The Generation Gap' by Benjamin M. Spock
   The Lotos Eaters – Alfred Tennyson, The Land of High Passes – Bhagyalakshmi Krishnamurthy
2. **Grammar:** Degrees of comparisons, Conjunctions, and Prepositions – Identifying Common Errors

3. **Vocabulary:** Word Formation (Affixation, Compounding, Conversion, Blending, Borrowing)
4. **Writing:** Punctuation, Clauses and Sentences, Transitional Devices, Paragraph Writing- Process

UNIT-II:
1. **Reading:** Emerging Technologies
   Medicate – Oleksandr Gorpynich Of Studies – Francis Bacon
2. **Grammar:** Articles Noun-Pronoun Agreement; Concord
3. **Vocabulary:** Word Formation (Prefixes, Suffixes, Root Words)
4. **Writing:** Essay Writing -Descriptive, Argumentative, Expository

UNIT-III:
1. **Reading: Leisure- William Henry Davies -Be Thankful - Unknown Author**
   The Rising of the Moon – Lady Gregory, With The Photographer – Stephen Leacock
2. **Grammar:** Misplaced Modifiers
3. **Vocabulary:** Synonyms and Antonyms
4. **Writing:** Letter Writing- Formal Letter Writing– Letter of Complaint, Letter of Requisition, Email Writing, Email Etiquette

UNIT-IV:
1. **Reading:** -Why a Start-Up Needs to Find its Customers First'- Pranav Jain
   Too dear – Leo Tolstoy, Shooting an Elephant – George Orwell
2. **Grammar:** Clichés, Redundancies
3. **Vocabulary:** Common Abbreviations
4. **Writing:** Summary Writing, Job Application, Resume Writing

UNIT-V:
1. Reading: **'Professional Ethics'**
2. Patterns of Writing: Comparison and Contrast
3. Patterns of Writing: Cause and Effect Paragraphs
4. Patterns of Writing: Classification Paragraph
5. Patterns of Writing: Problem–Solution Pattern of Writing

TEXT BOOKS:

1. Board of Editors,English for the Young in the Digital World, Orient Black Swan, 2025
2. Unlocking English: A Skill-Based Approach for Language Proficiency, Cengage, 2025
3. Technical Communication, Rebecca E. Burnett, 6th Edition, Cengage Learning

REFERENCES:

1. Practical English Usage, Swan Michael, New Edition, Oxford University Press, 2016
2. English Grammar Just for You, Karal, Rajeevan, Oxford University Press, 2023
3. Empowering with Language: Communicative English for Undergraduates. Cengage Learning India, 2024
4. Communication Skills – A Workbook, Sanjay Kumar & Pushp Lata, Oxford University Press, 2022
5. Study Writing, Hamp, Liz., Lyons and Heasly, Ben, C U P, 2006

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. I Semester

(25ES1ME101) ENGINEERING MECHANICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Basics of Physics & Mathematics

COURSE OBJECTIVES:
- To understand, analyze the forces and moment systems for equilibrium
- To know the concept of centroid and area moment of inertia about any axes
- To distinguish between statics and dynamics & kinematics and kinetics
- To understand the work-energy principle and impulse-momentum principles

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Analyze the systems using equilibrium conditions and apply the concepts of mechanics to engineering applications and friction bodies
CO-2: Determine the centroid of composite areas and moment of inertia of areas
CO-3: Solve the kinematics and kinetics problems
CO-4: Apply work-energy principle, impulse-momentum principle to solve engineering problems

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	3	2	2	-	1	-	-	-	2	3	3	-
CO-2	3	3	3	2	-	1	-	-	-	1	3	2	1
CO-3	3	3	2	2	-	1	-	-	-	2	3	2	1
CO-4	3	3	2	2	-	1	-	-	-	2	3	3	-

UNIT-I:
System of Forces: Basic concepts of mechanics-units, classification of mechanics, force, characteristics of force, types of forces, Principle of transmissibility. Resultant of coplanar system of forces- parallelogram law of forces, triangle law of forces, polygon law of forces, method of resolution of forces, equilibrant and resultant of coplanar concurrent forces-problems.
Equilibrium of Forces: Free Body Diagram, equilibrium conditions, Lami's theorem, equilibrium of roller- problems. Moment, couple, Varignon's principle, resultant of parallel forces- problems.

UNIT-II:
Friction: Types of Friction, laws of friction, coefficient of friction, angle of friction, angle of repose, cone of friction, limiting friction. Equilibrium of connected bodies on rough horizontal and inclined planes- problems. Screw jack and differential screw jack concept.

Centroid & Centre of Gravity: Introduction, Centroid, Centroids of lines, Standard areas and volumes, Centroids of composite sections, Centre of gravity of simple bodies, Pappu's Theorems-Problems.

UNIT-III:

Area Moment of Inertia: Introduction to area moment of inertia, polar moment of inertia, radius of gyration, parallel axis theorem, perpendicular axis theorem, moment of inertia of standard sections -problems.

Mass Moment of Inertia: Introduction to mass moment of inertia, radius of gyration, transfer formula for Mass moments of inertia. Derivations of mass moment of Inertia for simple sections- rectangular plate, ring, circular disc, slender rod, cylinder, sphere & cone.

UNIT-IV:

Kinematics of Particles: Kinematics of particles- rectilinear motion, curvilinear motion – Projectiles-Numerical questions on Projectile motion.

Kinetics of Particles: Kinetics of particles – Newton's Second Law, Dynamic equilibrium, Inertia force, D Alembert's Principle and its applications to connected bodies-Numerical questions on connected bodies.

UNIT-V:

Work–Energy: Work and Energy principle, Application of Work-Energy principle to-spring systems and connected bodies-problems.

Impulse–Momentum: Impulse-Momentum Principle, Application of Impulse-Momentum principle to connected bodies- Numerical questions on connected bodies.

TEXT BOOKS:

1. Engineering Mechanics, S. Timoshenko, D. H. Young & J. V. Rao, 5th Edition, TMH, 2017
2. Singer's Engineering Mechanics, K. Vijaya Kumar Reddy & J. Suresh Kumar, 3rd Edition, B. S. Publishers, 2024

REFERENCES:

1. Engineering Mechanics, J. L. Meriam & L. G. Kraige, 9th Edition, Wiley Publishers, 2020
2. Engineering Mechanics, R. C. Hibbeler, 15th Edition, Pearson Education, 2022
3. Engineering Mechanics, A. K. Tayal, 14th Edition, Umesh Publications, 2012
4. Engineering Mechanics, R. K. Rajput, 2nd Edition, Laxmi Publications, 2013
5. A Textbook of Engineering Mechanics, R. K. Bansal, 6th Edition, Laxmi Publications, 2023

ONLINE RESOURCES:

1. NPTEL – Engineering Mechanics, Prof. S. K. Mondal, IIT Roorkee
2. Coursera – Introduction to Engineering Mechanics (University of Georgia)

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. I Semester

(25PC1ME101) MANUFACTURING PROCESSES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:

- To understand sand casting and metal casting techniques
- To impart the knowledge of various welding processes
- To evaluate the importance of rolling, forging and sheet metal operations
- To analyse the processing of plastics

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Explain the steps and principles of casting processes and classify common casting defects
CO-2: Differentiate various welding processes and identify the causes of welding defects
CO-3: Apply the principles of rolling and sheet metal operations to produce simple components
CO-4: Explain extrusion and drawing processes and compare their applications
CO-5: Classify forging and plastic processing methods and describe their applications

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	2	2	-	-	-	1	-	-	-	-	-	2	1
CO-2	2	2	-	-	1	2	-	-	-	-	-	2	1
CO-3	2	-	1	-	-	1	-	-	-	-	-	2	1
CO-4	2	-	1	-	-	2	-	-	-	-	-	2	2
CO-5	2	-	-	-	-	2	-	-	-	-	-	2	1

UNIT-I:
Casting: Steps involved in making a casting; advantages of casting and its applications; types of foundry sands and their properties, Sand moulding, types of patterns – materials used for patterns, pattern allowances and their construction, principles of gating, Riser, Function, Types of Risers, Classification of Casting processes, Defects in castings - Causes and remedies.

UNIT-II:
Introduction: Classification of welding processes.
Arc Welding: Types of welds and welded joints, Welding Positions, Arc welding, shielded metal arc welding, Submerged arc welding, Friction Stir Welding, Resistance welding.
Gas Welding: Gas welding, types, tungsten inert gas welding, metal inert gas welding, soldering & brazing. welding defects, Causes and remedies.

UNIT-III:

Mechanical Working-I:

Hot Working and Cold Working: a comparison of properties between cold- and hot-worked parts, Strain hardening.

Sheet Metal Working: Stamping, forming and other cold working processes: Blanking and piercing; Bending and forming; Coining.

Rolling: Rolling fundamentals; theory of rolling; types of rolling mills and products.

UNIT-IV:

Mechanical Working-II:

Extrusion: Basic extrusion process and its characteristics; Hot extrusion and Cold extrusion; Forward extrusion and backward extrusion, Impact extrusion, Tube extrusion, hydrostatic extrusion.

Drawing: Principle of drawing and its types; Wire drawing and Tube drawing.

UNIT-V:

Forging Processes: Principle of forging; types of forging; smith forging; drop forging; roll forging; rotary forging, Forging defects.

Plastic Materials and Processes: Types of plastics; advantages of plastics, Injection moulding; Blow moulding; Thermoforming. Compression moulding.

TEXT BOOKS:

1. Manufacturing Technology: Foundry, Forming and Welding, P. N. Rao, Vol. 1, 5th Edition, McGraw-Hill Education, 2018
2. Manufacturing Engineering and Technology, Serope KalpakJain, Steven R. Schmid, 8th Edition, Pearson Publishers, 2023

REFERENCES:

1. A Textbook of Production Technology (Manufacturing Processes), Dr. P. C. Sharma, 11th Revised Edition, S. Chand, 2022
2. Production Technology, R. K. Jain, 19th Edition, Khanna Publication, 2010
3. Process and Materials of Manufacture, Roy A. Lindberg, 4th Edition, Prentice-Hall of India, 1990
4. Principles of Metal Casting, Richard W. Heine, Carl R. Loper, Jr., Philip C. Rosenthal, 3rd Edition, McGraw-Hill Education, 2015
5. Fundamentals of Modern Manufacturing: Materials, Processes, and Systems, Mikell P. Groover, 7th Edition, Wiley, 2023

ONLINE RESOURCES:

1. https://onlinecourses.nptel.ac.in/noc20_me67/preview
2. https://onlinecourses.nptel.ac.in/noc24_me48/preview

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. I Semester

(25BS2PH101) APPLIED PHYSICS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To explore the working principle of lasers and optical fibers through hands-on experiments
- To analyze the characteristics of semiconductors and devices
- To study resonance and magnetic induction in electromagnetic systems
- To synthesize the magnetic materials and study magnetic, dielectric properties of materials
- To develop skills in data analysis, interpretation, and scientific reporting

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Verify laser and optical fibre parameters through experimental study
CO-2: Characterize semiconductors and devices such as solar cell, laser diode etc
CO-3: Demonstrate resonance phenomena and realize tangent law of magnetism
CO-4: Understand the magnetic and dielectric properties of materials
CO-5: Apply scientific methods for accurate data collection, analysis, and technical report writing

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	3	1	2	1	2	1	3	2	-	1	1	-
CO-2	3	3	1	2	1	2	1	3	2	-	1	1	-
CO-3	3	3	1	2	1	2	1	3	2	-	1	1	-
CO-4	3	3	1	2	1	2	1	3	2	-	1	1	-
CO-5	3	3	1	2	2	2	1	3	3	2	1	-	2

LIST OF EXPERIMENTS:

1. a) Determination of wavelength of a laser using diffraction grating.
   b) Study of V-I & L-I characteristics of a given laser diode.
2. a) Determination of numerical aperture of a given optical fibre
   b) Determination of bending losses of a given optical fibre.
3. Synthesis of magnetite (Fe_3O_4) powder using sol-gel method.
4. V-I characteristics of solar cell.
5. Magnetic field along the axis of current carrying coil – Stewart and Gee
6. Melde’s Experiment.

7. Study of B-H curve of a ferro magnetic material.
8. Study of P-E loop of given ferroelectric crystal.
9. Determination of dielectric constant of a given material.
10. Determination of frequency of AC mains using sonometer.

(Note: Any 8 experiments are to be performed.)

TEXT BOOKS:

1. Engineering Physics Laboratory Manual/Observation, Physics Faculty, VNRVJIET
2. A Textbook of Practical Physics, S. Balasubramanian, M. N. Srinivasan, S. Chand Publishers, 2017

REFERENCES:

1. Engineering Physics Lab Manual, Dr. Y. Aparna & Dr. K. Venkateswarao, VGS Book
2. Engineering Physics Practicals, Dr. B. Srinivasa Rao, V. K. V. Krishna, K. S. Rudramamba, University Science Press, Imprint of Laxmi Publications

ONLINE RESOURCES:

1. https://vlab.amrita.edu/index.php?sub=1&brch=189&sim=343&cnt=1
2. https://vlab.amrita.edu/index.php?sub=1&brch=280&sim=1518&cnt=1

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. I Semester

(25HS2EN101) ENGLISH LANGUAGE AND COMMUNICATION SKILLS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

INTRODUCTION:
The aim of the English Language and Communication Skills Lab (ELCS Lab) is to develop the communicative skills of learners to function productively in academic and professional settings. This course focuses on developing Listening Skills, Reading Skills, Speaking Skills, Information transfer, Vocabulary, and pronunciation.

Professional students need reading strategies to explore scientific information and process the information. Further, they need to present and share this information in speaking and/or writing with their peers and scientific community. In addition to these, they also need good vocabulary for the presentation of their ideas. Practice is given not only in processing information to identify main ideas and relevant facts, to compare and contrast information and finally draw conclusions but also in presenting the same in speaking and writing.

COURSE OBJECTIVES:
- To train students to use neutral accent through phonetic sounds, symbols, stress and intonation
- To provide practice in vocabulary usage & grammatical construction
- To provide ample practice in LSRW skills and train the students in oral presentations, public speaking, role play, and situational dialogue
- To provide practice in defining technical terms and describing processes
- To equip students with excellent writing skills and information transfer skills

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Speak fluently with a neutral accent
CO-2: Use contextually apt vocabulary and sentence structures
CO-3: Make Presentations with great confidence
CO-4: Define technical terms and describe processes
CO-5: Write accurately, coherently, and lucidly

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	1	2	2	2	-	-	-	2	3	-	2	-	-
CO-2	1	2	2	2	3	3	2	2	3	2	2	2	2
CO-3	-	1	2	1	-	1	2	2	3	2	2	1	1
CO-4	2	2	2	2	2	-	-	1	3	2	2	2	2
CO-5	1	2	1	2	2	2	2	1	3	3	3	1	1

LIST OF EXPERIMENTS:

UNIT-I:

CALL

1. Common errors with Speech Sounds – Identify elements of Mother Tongue Influence,
2. Listening Skills – Types of Listening- Barriers – Active Listening Practice
3. Reading Skills -Skimming & Scanning-Reading for Gist -Reading for Specific Details

Oral Communication Lab:

1. Non-Verbal Communication
2. Formal and Informal English, Small Talk
3. Greetings – Introducing oneself & others - Elevator Pitch

UNIT-II:

CALL

1. **Activity:** Listening for information
2. Listening comprehension
3. **Reading Skills:** Reading comprehension-contextual vocabulary

Oral Communication Lab

1. Participatory Conversation
2. Role Playing in Formal Situations, Situational Dialogues – Two-Way (E. g. Elevator Pitch)
3. Media Etiquette, Conference Etiquette

UNIT-III:

CALL

1. **Phonetics:** Errors in Pronunciation, Neutralizing Mother Tongue Influence
2. **Listening Skills:** Listening for Gist
3. **Reading Skills:** SQ3R Method, Inferring

Oral Communication Lab

1. Describing Objects / Situations/ Places, People, Events,
2. Process Description

UNIT-IV:

CALL

1. **Listening Skills:** Techniques for Listening –Listening for Specific Details
2. **Reading Skills:** Making inferences

Oral Communication Lab

1. Story Telling; Story STAR, Sequencing, Creativity
2. Telling and Retelling Stories- Collage

UNIT-V:

CALL

1. **Listening Skills:** Listening for Evaluation, Writing the Summary
2. **Prompt Engineering:** Generating Appropriate Prompts, Keyword Extraction
3. Definition of a Technical Term; System Flow

Oral Communication Lab

1. Report Presentation
2. Short Presentation

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. I Semester

(25PC2ME101) MANUFACTURING PROCESSES LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:
- To provide hands-on experience with casting, welding, forming, and machining operations
- To help students understand the working principles of manufacturing processes
- To expose students to tools, machines, and equipment used in production workshops
- To develop safety awareness and quality control in manufacturing practices
- To bridge theoretical knowledge with practical application of manufacturing methods

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Demonstrate casting practices such as pattern making and sand moulding
CO-2: Perform welding operations like arc welding, gas welding, and spot welding
CO-3: Understand sheet metal fabrication, bending, and joining processes
CO-4: Demonstrate processing of plastics by injection moulding and blow moulding
CO-5: Practice safety, accuracy, and productivity in production environments

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	1	-	-	1	-	-	-	2	3	-	2	1	-
CO-2	2	-	2	1	-	-	-	2	1	1	2	3	2
CO-3	1	-	2	1	-	-	-	2	1	-	2	1	-
CO-4	1	-		1	-	-	-	2	3	-	2	2	-
CO-5	1	-	-	1	-	-	1	1	3	1	3	2	-

LIST OF EXPERIMENTS:

Experiments to be performed from the following:

I. Metal Casting:
1. Pattern Design and making
2. Sand properties testing
3. Moulding, Melting and Casting

II. Welding:
1. Lap and Butt Joint using ARC Welding -2 Exercises

2. Spot Welding
3. TIG Welding
4. Gas Welding and Brazing-2 Exercises

III. Mechanical Working:

1. Fly Press: Blanking and Piercing operations using Fly press
2. Hydraulic Press: Bending operation

IV. Processing of Plastics:

1. Injection Molding
2. Blow molding

Note: A Minimum of 10 Exercises needs to be performed

TEXT BOOKS:

1. Manufacturing Technology Volume-I, P. N. Rao, McGraw-Hill Higher Education
2. Production Technology, R. K. Jain, Khanna Publishers.

REFERENCES:

1. Principles of Metal Castings, Rosenthal, Publisher, McGraw Hill Education
2. Welding Processing and Technology, R. S. Parmar, 2nd Edition, Khanna Publishers
3. Production Technology, Sharma P. C., S. Chand

ONLINE RESOURCES:

1. https://onlinecourses.nptel.ac.in/noc21_me99/preview
2. https://onlinecourses.nptel.ac.in/noc21_me23/preview

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. I Semester

(25ES2ME101) ENGINEERING WORKSHOP

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To know the different popular manufacturing process
- To gain a good basic working knowledge required for the production of various engineering products
- To provide hands on experience about use of different engineering materials, tools, equipment and processes those are common in the engineering field
- To identify and use marking out tools, hand tools, measuring equipment and to work to prescribed tolerances

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand various types of manufacturing processes

CO-2: Fabricate/make components from wood and steels through hands on experience

CO-3: Understand different machining processes like turning, drilling, tapping, etc.

CO-4: Understand electrical and electronic components and their assembly

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	-	-	1	-	-	-	-	3	3	2	-	2	1
CO-2	-	-	1	-	-	-	-	3	3	2	-	2	1
CO-3	-	-	1	-	-	-	-	3	3	2	-	2	1
CO-4	-	-	1	-	-	-	-	3	3	2	-	1	1

LIST OF EXPERIMENTS:

1. TRADES FOR EXERCISES:

i) **Carpentry:** Cross lap joint, Mortise & tenon joint
ii) **Fitting:** L-fitting, Square fitting
iii) **Tin smithy:** Rectangular tray, Rectangular scoop
iv) **Arc welding:** Lap joint, Butt joint
v) **House Wiring:** Single lamp connection, Staircase connection
vi) **Black smithy:** Round to Square, U-Hook
vii) **Machine shop:** Step turning on lathe, Drilling & tapping

2. TRADES FOR DEMONSTRATION:

Plumbing, 3D printing, Power tools in construction and wood working.

TEXT BOOKS:

1. Workshop Manual, P. Kannaiah and K. L. Narayana, 3rd Edition, Scitech, 2015
2. Elements of Workshop Technology Vol. 1 & 2, S. K. Hajra Choudhury, A. K. Hajra Choudhury and Nirjhar Roy, 13th Edition, Media Promoters & Publishers, 2010

REFERENCES:

1. Manufacturing Engineering and Technology, Serope Kalpakjian, Steven R. Schmid, 4th Edition, Pearson Education India Edition, 2002
2. Manufacturing Technology-I, S. Gowri, P. Hariharan and A. Suresh Babu, Pearson Education, 2008
3. Processes and Materials of Manufacture, Roy A. Lindberg, 4th Edition, Prentice Hall India, 1998
4. Manufacturing Technology Vol-1 & 2, P. N. Rao, Tata McGraw-Hill, 2017

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. II Semester

(25BS1MT105) ORDINARY DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS
(Common to CE, ME and AE)

TEACHING SCHEME		
L	T/P	C
2	1	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Mathematical Knowledge at Pre-University level

COURSE OBJECTIVES:
- To learn methods of solving first order differential equations and learn about its applications to basic engineering problems.
- To learn methods of solving higher order differential equations with constant coefficients and learn about its applications to basic engineering problems.
- To learn methods of solving higher order differential equations with variable coefficients and learn about its applications to basic engineering problems.
- To learn the physical quantities involved in engineering field related to vector valued functions
- To know the basic properties of vector valued functions and their applications to line, surface and volume integrals

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Formulate and solve the problems of first order differential equations
CO-2: Solve higher order differential equation with constant coefficients and apply the concept of differential equation to real world problems
CO-3: Solve the higher order differential equation with variable coefficients and apply the concept of differential equation to real world problems
CO-4: Find the gradient, divergence, curl and its physical interpretations
CO-5: Evaluate the line, surface and volume integrals and converting them from one to another

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	3	1	2	1	-	-	-	-	-	-	2	1
CO-2	3	3	1	2	1	--	-	-	-	-	-	2	1
CO-3	3	2	1	2	2	-	-	-	-	-	-	2	2
CO-4	3	2	2	2	1	-	-	-	-	-	-	1	2
CO-5	3	2	2	3	1	-	-	-	-	-	-	2	1

UNIT-I:
First Order Ordinary Differential Equations: Exact differential equations – Equations reducible to exact differential equations – linear and Bernoulli's equations – Orthogonal Trajectories (only in Cartesian Coordinates). Applications: Newton's law of cooling – Law of natural growth and decay.

UNIT-II:
Higher Order ODEs with Constant Coefficients: Higher order linear differential equations with constant coefficients: Non-Homogeneous terms of the type e^{ax}, sin ax, cos ax, polynomials in x, e^{ax} V(x) and x V(x).

UNIT-III:
Higher Order ODEs with Variable Coefficients: Higher order linear Ordinary differential equations with variable coefficients, Method of variation of parameters, Euler-Cauchy ODEs, Legendre's ODEs.

UNIT-IV:
Vector Differentiation: Vector point functions and scalar point functions – Gradient – Directional derivatives -Divergence and Curl - Solenoidal and Irrotational vectors - Scalar potential functions – Vector Identities (without proof).

UNIT-V:
Vector Integration: Line, Surface and Volume Integrals. Theorems of Green, Gauss and Stokes (without proofs) and their applications.

TEXT BOOKS:

1. Higher Engineering Mathematics, B. S. Grewal, 36th Edition, Khanna Publishers, 2010
2. Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, 5th Editon, Narosa Publications, 2016
3. Higher Engineering Mathematics, B. V. Ramana, 11th Reprint, Tata McGraw-Hill, 2010

REFERENCES:

1. Advanced Engineering Mathematics, Erwin Kreyszig, 9th Edition, John Wiley & Sons, 2006
2. Advanced Engineering Mathematics, Dennis G. Zill, 6th Edition
3. A Text Book of Engineering Mathematics, N. P. Bali and Manish Goyal, Laxmi Publications, Reprint, 2008
4. Higher Engineering Mathematics, H. K. Dass and Er. Rajnish Verma, S. Chand

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. II Semester

(25BS1CH102) ENGINEERING CHEMISTRY

(Common to CE, ME and AE)

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: General Chemistry and Basic Mathematics

COURSE OBJECTIVES:
- To analyze the quality of water for sustainable living
- To imbibe the conceptual knowledge of electrochemical processes and corrosion science
- To outline the importance of non-conventional energy sources and portable electric devices
- To acquire the knowledge about polymer science and its applications in various fields
- To recognize the significance of analytical techniques and materials in engineering, industrial, environmental, and biomedical applications

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Assess the specification of water regarding its usage in domestic & Industrial scenarios
CO-2: Interpret electrode potentials and predict the suitable corrosion control methods in safeguarding the structures
CO-3: Recognize the transformations in energy sources & battery technology
CO-4: Analyze the efficacy of polymers in diverse applications
CO-5: Identify the principles of UV-Visible and IR spectroscopy and materials in varied applications

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	2	2	2	2	2	1	1	1	1	2	1	1
CO-2	3	2	2	1	1	2	1	1	1	1	2	1	1
CO-3	3	2	2	1	1	1	1	1	1	1	2	1	1
CO-4	3	2	2	1	1	2	1	1	1	1	2	1	1
CO-5	3	2	1	1	1	1	1	1	1	1	2	1	1

UNIT-I:
Water and its Treatment: Introduction – Hardness, types, degree of hardness and units – numerical Problems. Estimation of hardness of water by complexometric method. Potable water and its specifications (WHO) – Steps involved in the treatment of potable water – screening, sedimentation, coagulation, filtration and disinfection by chlorination and break-point chlorination.

Boiler Troubles: Scales, Sludges, Internal treatment of boiler feed water – Calgon conditioning, Phosphate conditioning, Colloidal conditioning. External treatment methods – Softening of water by Ion-exchange process. Membrane separation – Types of membranes [Microfiltration (MF), Ultrafiltration (UF), Nanofiltration (NF)], advantages and applications. Desalination of brackish water – Reverse osmosis.

UNIT-II:

Electrochemistry and Corrosion: Introduction – Electrochemical cell & galvanic cell, electrode potential, standard electrode potential, Nernst equation (no derivation). Types of electrodes – Reference electrodes – Construction, working principle, advantages, limitations and applications of Primary reference electrode (Standard Hydrogen Electrode – SHE), Secondary reference electrode (Calomel electrode) and Ion-selective electrode (Combined glass electrode).

Corrosion: Introduction – Definition, causes and effects of corrosion, Theories of corrosion – chemical and electrochemical theories of corrosion. Types of corrosion – galvanic, waterline and pitting corrosion. Factors affecting rate of corrosion – nature of metal (position, passivity, purity, areas of anode and cathode) & nature of environment (temperature, pH, humidity). Corrosion control methods – Cathodic protection Method – Sacrificial anode and impressed current cathodic protection methods.

UNIT-III:

Energy Sources and Battery Technology:

Fuels: Introduction and characteristics of good fuel, classification based on physical state.

Fossil Fuels: Petroleum – Refining of Crude oil, Cracking – Types of cracking – Moving bed catalytic cracking. Synthetic Fuels: Fischer-Tropsch process.

Gaseous Fuels: Composition and uses of LPG, CNG and Hythane. Green Hydrogen (generation by water electrolysis – PEM, Alkaline, Solid Oxide).

Batteries: Introduction – Classification of batteries – Primary, secondary, reserve batteries and fuel cells (differences and examples). Construction, working and applications of Zn-air (primary battery) and Lithium-ion battery (secondary battery), Characteristic features and Advantages of Metal ion batteries. (Sodium Ion Battery). Fuel Cells – Construction and applications of Proton Exchange Membrane Fuel Cell (PEMFC).

UNIT-IV:

Polymers: Definition – Classification of polymers – Types of polymerizations – Addition and condensation polymerization, differences between thermoplastics and thermosetting plastics. Plastics, Elastomers and Fibers: Synthesis, properties and applications of Teflon, Buna-S and Kevlar.

Fiber Reinforced Plastics (FRPs): Classification, features and applications.

Conducting Polymers: Definition, classification and applications.

Shape Memory Polymers: Definition, classification and applications.

Biodegradable polymers: Introduction – Definition features & applications of Polylactic acid (PLA).

UNIT-V:

Spectroscopic Applications and Engineering Materials: Interpretative spectroscopic applications of UV-Visible spectroscopy for analysis of pollutants in dye industry, IR spectroscopy in night vision – security, Pollution Under Control – CO sensor (Passive Infrared detection).

Self-healing Materials: Definition, working principle and applications (self-healing concrete).
Nanomaterials: Characterization by STM & AFM.
Lubricants: Definition and characteristics of a good lubricant – thin film mechanism of lubrication, properties of lubricants – viscosity, cloud and pour point, flash and fire point.

TEXT BOOKS:

1. Engineering Chemistry, P. C. Jain and M. Jain, Dhanpat Rai and Company, 2010
2. Engineering Chemistry, Rama Devi, Dr. P. Aparna and Rath, Cengage learning, 2025
3. Engineering Chemistry, Thirumala Chary, Laxminarayana & Shashikala, Pearson Publications, 2020

REFERENCES:

1. Engineering Chemistry, Shashi Chawla, Dhanpat Rai and Company, 2011
2. Engineering Chemistry, Shikha Agarwal, Cambridge University Press, 2015
3. Textbook of Engineering Chemistry, Jaya Shree Anireddy, Wiley Publications
4. Engineering Analysis of Smart Material Systems, Donald J. Leo, Wiley, 2007
5. Challenges and Opportunities in Green Hydrogen, Paramvir Singh, Avinash Kumar Agarwal, Anupma Thakur, R. K. Sinha, Springer Nature, 2024

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. II Semester

(25ES1EE104) ELEMENTS OF ELECTRICAL AND ELECTRONICS ENGINEERING

TEACHING SCHEME		
L	T/P	C
2	0	2

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Mathematics, Physics

COURSE OBJECTIVES:

- To understand the use of electrical energy in different engineering fields
- To analyze electrical circuits using different network reduction techniques
- To practice the techniques to control, assess electrical machines and installations
- To learn principles of operation, construction and characteristics of various electronic devices

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Appreciate the role of electrical energy in various engineering branches and analyze the DC circuits using various network reduction techniques
CO-2: Analyze the various electrical parameters of AC circuits with R-L-C elements
CO-3: Understand the operation of various Electrical Machines & Installations
CO-4: Apply basic electronics device in real time applications

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	2	1	1	-	-	-	-	-	-	1	2	-
CO-2	2	2	1	1	-	-	-	-	-	-	1	2	-
CO-3	2	1	1	1	-	1	-	-	-	-	1	2	-
CO-4	2	1	1	1	-	-	-	-	-	-	1	2	-

UNIT-I:

Introduction to Electrical Energy and DC Circuits: The role of Electrical Energy in modern life and various engineering branches- Circuit Concept – Types of Elements R-L-C parameters – Voltage and Current sources – Independent and dependent sources- Kirchhoff's laws – network reduction techniques: series, parallel, series parallel, star/delta transformations, Superposition theorem

UNIT-II:

Steady State AC Circuits: Representation of sinusoidal waveforms, average and RMS values, form factor and peak factor, phasor representation, Analysis of single-phase AC circuits consisting of R, L, C, series RL, RC, RLC combinations, real power, reactive power, apparent power, power factor - Three-phase balanced circuits, voltage and current relations in star and delta connections (Derivation only).

UNIT-III:

Electrical Machines: Transformer principle, types, Equivalent circuit, Regulation and Efficiency, DC generator principle, types Emf equation, DC motor principle, types, Back emf, Speed control of shun motor. Three phase induction motor, types, principle, torque Slip characteristics, working principle of Synchronous generator, principle of operation of synchronous motor.

UNIT-IV:

Electrical Installations:

Components of LT Switch Gear: SFU, MCB, ELCR, MCCB. Types of wiring, Need of earthing and types. Batteries-types-working and electrical characteristics of Lead-Acid battery, Elementary energy consumption calculations, Block diagram of UPS and its operation

UNIT-V:

Diode & Bipolar Junction Transistor: PN Junction Diode, Operation and Symbol, V-I Characteristics, Half Wave Rectifier, Full-Wave Rectifier, (Centre Tap and Bridge Rectifiers), Zener Diode, V-I Characteristics and applications, BJT, Construction (NPN and PNP transistors), Transistor configurations (CC, CB and CE), Transistor biasing, Transistor as an Amplifier.

TEXT BOOKS:

1. Basic Electrical Engineering, D. C. Kulshreshtha, 2nd Edition, TMH, 2019
2. Basic Electrical Engineering, P. Ramana, M. Suryakalavathi, G. T. Chandra Sekhar, 1st Edition, S. Chand, 2018
3. Electronic Devices and Circuits, S. Salivahanan and N. Suresh Kumar, 3rd Edition TMH, 2019

REFERENCES:

1. Electrical and Electronics Technology, E. Hughes, 10th Edition, Pearson, 2010
2. Electrical Engineering Fundamentals, Vincent Deltoro, 2nd Edition, Prentice Hall India, 1989
3. Electrical and Electronics Measurements and Instrumentation, A. K. Sawhney, 3rd Edition, Dhanpat Rai & Co., 1983
4. Basic Electrical Engineering, D. P. Kothari and I. J. Nagrath, 3rd Edition, Tata McGraw-Hill, 2010
5. Engineering Circuit Analysis, William Hayt and Jack E. Kemmerly, 8th Edition, McGraw-Hill, 2013

ONLINE RESOURCE:

1. https://nptel.ac.in/courses/117106108

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. II Semester

(25ES1AM101) C PROGRAMMING AND DATA STRUCUTRES

(Common to CE and ME)

TEACHING SCHEME		
L	T/P	C
2	0	2

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:

- To understand the fundamentals of C language and apply decision-making statements for developing simple programs
- To learn iteration, arrays, and string handling in C for problem solving
- To learn sorting, searching, and function concepts in C for efficient problem solving
- To understand structures, unions, and pointers with dynamic memory allocation in C programming
- To understand and analyze various data structures

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Explore the basic concepts in C Programming Language
CO-2: Apply loops, arrays, and string functions to solve real world problems
CO-3: Develop modular programs using different language constructs and analyze various searching and sorting algorithms
CO-4: Demonstrate the ability to declare, define, and manipulate structures, unions, and derived types for solving programming problems
CO-5: Apply data structures such as stacks, queues in problem solving

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	1	2	2	-	-	-	-	1	1	1	1	-	-
CO-2	3	2	3	1	2	-	-	-	1	-	-	-	-
CO-3	3	3	3	2	2	-	-	1	1	1	1	1	1
CO-4	3	3	3	2	-	-	-	-	1	-	-	-	-
CO-5	3	3	3	2	2	-	-	2	2	1	1	-	-

UNIT-I:
Introduction to C Language: Algorithm, Flowchart, Structure of C program, Identifiers, Basic data types, Variables, Constants, Input / Output Statements, Operators, Expressions, Precedence and Associatively, Expression Evaluation, Type conversions statements.
Statements: If and switch statements- Example C Programs.

UNIT-II:
Repetition Statements: while, for, do-while statements, break, continue, goto**.**
Arrays: Basic concepts- one-dimensional and two-dimensional arrays- initialization and declaration.
Character Array: string handling functions - example C programs.

UNIT-III:

Searching: Linear and Binary search methods.

Sorting: Bubble sort- Selection sort-Insertion sort.

Functions: Functions, basics, user defined functions, inter function communication, storage classes, scope rules

Recursive Functions: factorial and Fibonacci series example C Programs.

UNIT-IV:

Structures: Declaration-Definition and Initialization of Structures-Accessing Structures-Operations on Structures.

Unions: Declaration-Definition and Initialization of Unions**.**

Pointers: Basic concepts- Pointers and arrays- Pointers and strings - Pointers and structures. Self-referential structures, Dynamic Memory Allocation- Example C programs.

UNIT-V:

Introduction to Data Structures: Abstract data types.

Stacks: Operations, array and linked representations of stacks

Queues: Operations, array and linked representations.

Linear List: singly linked list implementation, insertion, deletion and searching operations on linear list.

Trees and Graphs: Definition and representation.

TEXT BOOKS:

1. C Programming & Data Structures, B. A. Forouzan and R. F. Gilberg, 3rd Edition, Cengage Learning
2. Problem Solving and Program Design in C, J. R. Hanly and E. B. Koffman, 5th Edition, Pearson Education
3. The C Programming Language, B. W. Kernighan and Dennis M. Ritchie, PHI/Pearson Education

REFERENCES:

1. C & Data Structures, P. Padmanabham, 3rd Edition, B. S. Publications
2. C Programming with Problem Solving, J. A. Jones & K. Harrow, Dreamtech Press
3. C Programming & Data Structures, E. Balagurusamy, TMH
4. C Programming & Data Structures, P. Dey, M Ghosh R Thereja, Oxford University Press

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. II Semester

(25PC1ME102) THERMODYNAMICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Physics, Mathematics

COURSE OBJECTIVES:
- To apply the basic concepts of thermodynamics, heat and work done on the system
- To apply the basic concepts of Thermodynamic Laws for various thermodynamic systems
- To evaluate the properties of pure substance and to analyse the concept of irreversibility and availability
- To apply the basic concept of power cycles for External combustion engines and internal combustion engines
- To evaluate the behaviour of ideal gas mixtures and thermodynamic properties

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Apply the first law of thermodynamics to systems for energy balance and enthalpy evaluation
CO-2: Analyze heat engines, refrigerators, and heat pumps using the second law, entropy and Carnot principles
CO-3: Interpret thermodynamic property data and diagrams to analyze phase change and evaluate power cycle efficiency
CO-4: Determine thermodynamic properties of gas mixtures using Dalton's law, specific heats, Maxwell relations, and T-ds equations

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	2	1	-	-	-	-	-	-	-	-	2	-
CO-2	3	2	1	-	-	-	-	-	-	-	-	2	-
CO-3	3	2	1	-	-	-	-	-	-	-	-	2	-
CO-4	3	2	1	-	-	-	-	-	-	-	-	2	-

UNIT-I:
Concepts and Definitions: Thermodynamic system and control volume; Macroscopic versus microscopic point of view; Properties and state of a substance; Processes and cycles, Energy, Specific volume and density, The Zeroth law of thermodynamics; Temperature scales.
Work and Heat: Work transfer; displacement work; path and point function; pdv work in various Quasi-Static processes; Free expansion with zero work transfer, Heat transfer, Heat transfer a path function; Comparison of heat and work.

UNIT-II:

The First Law of Thermodynamics: The first law of thermodynamics for a closed system undergoing a cycle; First law for a closed system undergoing a change of state; Energy-a property of system; Different forms of stored energy; specific and latent heat; Specific heat at constant volume; Enthalpy; Specific heat at constant pressure; Perpetual motion machine of the first kind-PMM1

First Law Analysis for a Control Volume: Conversion of mass and the control volume, the first law of thermodynamics for a control volume, The steady-state process; Examples of steady-state processes.

UNIT-III:

The Second Law of Thermodynamics: Necessity of the Second Law of thermodynamics; Kelvin–Planck and Clausius Statements, Heat Engines, Efficiency of heat engines, Refrigerators and heat pumps, COP of refrigerators and heat pumps. The reversible process; Factors that render processes irreversible; The Carnot cycle; Two propositions regarding the efficiency of a Carnot cycle; The thermodynamic temperature scale; The ideal-gas temperature scale; Ideal versus real machines.

Entropy for a Control Mass: Clausius theorem; Entropy — a property of a system; Temperature entropy plot; The inequality of Clausius; Entropy change in irreversible process, Entropy principle; Mixing of two fluids; Entropy and direction.

UNIT-IV:

Properties of a Pure Substance: P-V diagram for a pure substance; P-T diagram for a pure substance; T-S diagram for a pure substance; Mollier diagram for a pure substance; Dryness fraction; Steam tables; saturated state; Liquid- vapor state; Superheated state.

Power Cycles: Rankine cycle; simple Brayton cycle; Otto cycle; Diesel cycle; Dual cycle, Description and representation on P-V and T-S diagram, Thermal Efficiency.

UNIT-V:

Properties of Gases and Gas Mixtures: Avogadro's Law; Ideal Gas; Equation of State; Properties of Mixture of Gases-Dalton's Law of Partial Pressures; Mole fraction, Volume fraction and partial pressure, Internal Energy, Enthalpy, and Specific Heats of Gas Mixtures; The Maxwell relations; Tds equations; Joule-Thompson coefficient.

TEXT BOOKS:

1. Engineering Thermodynamics, P. K. Nag, McGraw-Hill, 2017
2. Fundamentals of Thermodynamics, C. Borgnakke, R. E. Sonntag, and G. J. Van Wylen, John Wiley, 2020
3. Thermodynamics, An Engineering Approach, Yunus Cengel and Boles, TMH, 2019

REFERENCES:

1. Fundamentals of Engineering Thermodynamics, Moran M. J., Shapiro H. N., Boettner D. D., and Bailey M. B., John Wiley & Sons 2018
2. Engineering Thermodynamics, P. Chattopadhyay, Oxford University Press, 2015

ONLINE RESOURCES:

1. https://onlinecourses.nptel.ac.in/noc23_me76/preview
2. https://nptel.ac.in/courses/112105123

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. II Semester

(25ES3ME102) ENGINEERING GRAPHICS

TEACHING SCHEME		
L	T/P	C
0	6	3

EVALUATION SCHEME				
D-D	SE	CP	SEE	TOTAL
10	20	10	60	100

COURSE OBJECTIVES:
- To know the importance of engineering scales and curves
- To learn to use the orthographic projections for points, lines and planes in different positions
- To learn to draw the projections of solids and its sections in different positions
- To learn to draw the development of surfaces and intersection of solids
- To learn to draw the isometric projections of solids

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Apply the concepts of scales and engineering curves in construction using AutoCAD
CO-2: Solve the problem of projections of points, lines and planes in different positions using AutoCAD
CO-3: Draw the orthographic projections of solids and its sections in different positions using AutoCAD
CO-4: Obtain the development of surfaces of solids and intersection of solids using AutoCAD
CO-5: Construct the isometric projections of solids using AutoCAD

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	2	-	-	3	2	-	-	-	-	2	-	-
CO-2	3	2	-	-	3	2	-	-	-	-	2	-	-
CO-3	3	2	-	-	3	2	-	-	-	-	2	-	-
CO-4	3	2	-	-	3	2	-	-	-	-	2	-	-
CO-5	3	2	-	-	3	2	-	-	-	-	2	-	-

Introduction to AutoCAD Software: User interface, Menu System, Toolbars (Draw, Modify, Dimension, Layers), Setting drawing area, Status Bar (ortho, grid, snap, osnap, iso, lineweight), Display control commands (pan, zoom), Print setup.

UNIT-I:
Introduction to Engineering Graphics: Principles of Engineering Graphics and their Significance, Conventions, Drawing instruments.
Engineering Curves

Conic Sections: Ellipse, Parabola and Hyperbola- General methods only, Rectangular Hyperbola
Cycloidal Curves & Involutes: Cycloid, Epicycloid, Hypocycloid and Involutes
Scales: Plain, Diagonal and Vernier Scales

UNIT-II:
Orthographic Projections: Principles of Orthographic Projections, Conventions, Projections of Points in all positions; Projections of lines and planes inclined to both the planes - Auxiliary Views

UNIT-III:
Projections of Regular Solids: Projections of regular Solids like Prism, Cylinder, Pyramid and Cone inclined to both the Planes - Auxiliary Views
Sections of Solids: Section and sectional views of right regular solids like Prism, Cylinder, Pyramid and Cone – Auxiliary Views

UNIT-IV:
Development of Surfaces of Right Regular Solids: Development of surfaces of Right Regular Solids like Prism, Pyramid, Cylinder and Cone
Intersection of Solids: Intersection of prism vs prism, cylinder vs cylinder

UNIT-V:
Isometric Projections: Principles of isometric projection, Isometric Scale, Isometric Views, Conventions, Isometric Views of lines, Planes and Solids like Prism, Pyramid, Cylinder and Cone
Conversion of Isometric Views to Orthographic Views and Vice-versa, Conventions.

TEXT BOOKS:

1. Engineering Drawing, N. D. Bhatt, 53rd Edition, Charotar Publishing House, 2016
2. Textbook on Engineering Drawing, K. L. Narayana & P. Kannaiah, Scitech Publishers, 2010
3. Engineering Drawing and Computer Graphics, M. B. Shah & B. C. Rana, Pearson Education, 2010

REFERENCES:

1. Engineering Drawing and Graphics using AutoCAD, T. Jeyapoovan, 3rd Edition, S. Chand and Company
2. Engineering Drawing, Basant Agrawal and C. M. Agrawal, 3rd Editon, McGraw-Hill
3. Mastering AutoCAD 2021 and AutoCAD LT 2021, George Omura and Brian C. Benton (AutoCAD 2021), 1st Edition, John Wiley & Sons, Indianapolis, Indiana

ONLINE RESOURCES:
1. https://nptel.ac.in/courses/112103019
2. http://vlabs.iitb.ac.in/vlabs-dev/labs/mit_bootcamp/egraphics_lab/labs/index.php

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. II Semester

(25BS2CH101) ENGINEERING CHEMISTRY LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Basic Knowledge of Volumetric Analysis and Mathematics

COURSE OBJECTIVES:
- To understand the preparation of standard solutions and handling of instruments
- To determine and evaluate the water quality
- To measure physical properties like absorption of light, surface tension, pH, conductance, and viscosity of various liquids
- To conduct and collect the experimental data using different laboratory techniques
- To summarize the data and find the applicability to real world scenario

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Learn and apply the basic laboratory methodologies for the preparation of the standard solutions and handling of instruments
CO-2: Estimate the ions / metal ions present in domestic and industrial water
CO-3: Utilize the instrumental techniques to assess the physical properties of oils and water
CO-4: Analyze the experimental data to predict solutions for complex engineering problems
CO-5: Apply the skills gained to address societal issues related to real world scenario

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	2	1	1	-	1	1	1	1	1	2	-	1
CO-2	3	2	2	2	1	2	1	1	1	1	2	1	1
CO-3	3	2	1	2	1	2	1	1	1	1	2	1	1
CO-4	3	2	2	2	1	2	1	1	1	1	2	1	1
CO-5	3	2	2	2	1	2	1	1	1	1	2	1	1

LIST OF EXPERIMENTS:

VOLUMETRIC ANALYSIS:
1. Estimation of hardness of water by complexometric method using EDTA.
2. Determination of chloride content in the given sample water using Argentometric method.
3. Determination of Iron using Potassium Dichromate.

COLORIMETRY:

4. Estimation of copper present in the given solution by colorimetric method.

CONDUCTOMETRY:

5. Conductometric titration of Acid vs Base.
6. Conductometric titration of mixture of strong acid and weak acid by strong base.

pH METRY:

7. Titration of Acid vs Base using pH metric method.
8. Determination of viscosity of sample oil by Redwood Viscometer-I.
9. Estimation of acid value of given lubricant oil.
10. Determination of surface tension of a liquid by drop method using Stalagmometer.

PREPARATIONS:

11. Preparation of Bakelite.
12. Preparation of Nylon-6,6.
13. Determination of rate of corrosion of mild steel in the presence and absence of inhibitor.
14. **VIRTUAL LAB EXPERIMENTS / CBP:**

- Construction of advanced Fuel cell and it's working.
- Smart materials for Biomedical applications
- Batteries for electrical vehicles.
- Functioning of solar cell (materials used) and its applications.

TEXT BOOKS:

1. Laboratory Manual on Engineering Chemistry, S. K. Bhasin and Sudha Rani, Dhanpat Rai Publications
2. College Practical Chemistry, V. K. Ahluwalia, Sunitha Dhingra, Adargh Gulati, University Press
3. Practical Chemistry, Dr. O. P. Pandey, D. N. Bajpai, and Dr. S. Giri, S. Chand

REFERENCES:

1. Vogel's Textbook of Quantitative Chemical Analysis, G. N. Jeffery, J. Bassett, J. Mendham and R. C. Denny, Longmann, ELBS
2. Advanced Practical Physical Chemistry, J. D. Yadav, Goel Publishing House
3. Practical Physical Chemistry, B. D. Khosla, R. Chand and Sons
4. Lab Manual for Engineering Chemistry, B. Ramadevi and P. Aparna, S. Chand, 2022
5. Vogel's Textbook of Practical Organic Chemistry, 5th Edition

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. II Semester

(25ES2AM101) C PROGRAMMING AND DATA STRUCUTRES LABORATORY

(Common to CE and ME)

TEACHING SCHEME		
L	T/P	C
0	2	2

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To familiarize with the basic concepts of C programming through simple input/output, operators, and type conversion programs
- To develop problem-solving skills using decision-making, iteration, and control statements in C
- To provide practical knowledge of arrays, strings, searching, sorting, and function implementations
- To strengthen understanding of advanced concepts like recursion, structures, unions, and pointers
- To introduce the implementation of fundamental data structures such as stacks and queues using C

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Write, compile, and execute C programs using operators, type conversions, and input/output statements

CO-2: Apply decision-making and iterative constructs to solve computational problems

CO-3: Implement arrays, strings, searching, sorting, and functions effectively in C

CO-4: Develop programs using recursion, structures, unions, and pointers for real-time applications

CO-5: Design and implement stack and queue operations using C for basic data handling

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	2	2	-	-	-	-	-	-	-	-	-	-	-
CO-2	2	2	2	1	-	-	-	-	-	-	-	-	-
CO-3	2	2	1	2	1	-	-	-	2	1	-	-	-
CO-4	2	2	1	2	-	-	-	-	2	1	-	1	1
CO-5	2	2	2	2	1	-	-	2	2	1	-	1	1

LIST OF PROGRAMS / EXPERIMENTS / EXCERCISES:

WEEK-1:

C Programs on input and output statements

WEEK-2:
C Programs on operators
C Programs using type conversions

WEEK-3:
C Programs using if statements (Simple if, if-else, nested if else, else if ladder)
C Programs on switch statement.

WEEK-4:
C Programs using for, while and do-while loops
C Programs to illustrate break and continue statements.

WEEK-5:
C Programs to illustrate goto statement.
C Programs on 1-D, 2-D arrays.

WEEK-6:
C Programs to illustrate string handling functions.
C Programs on linear search and binary search

WEEK-7:
LAB INTERNAL -I

WEEK-8:
C Programs to illustrate Bubble sort, Selection sort and Insertion sort.
C Programs on functions.

WEEK-9:
C Program to find factorial of a given number using recursion
C Program to find Fibonacci series using recursion.

WEEK-10:
C Programs using structures and Unions

WEEK-11:
C Programs using pointers

WEEK-12:
C Program to implement stack using arrays.
C Program to implement queue using arrays

WEEK-13:
C Program to implement Single linked list.

WEEK-14:

LAB INTERNAL -II

TEXT BOOKS:

1. C Programming & Data Structures, B. A. Forouzan and R. F. Gilberg, 3rd Edition, Cengage Learning
2. Problem Solving and Program Design in C, J. R. Hanly and E. B. Koffman, 5th Edition, Pearson Education
3. The C Programming Language, B. W. Kernighan and Dennis M. Ritchie, PHI/Pearson Education

REFERENCES:

1. C & Data Structures, P. Padmanabham, 3rd Edition, B. S. Publications
2. C Programming with Problem Solving, J. A. Jones & K. Harrow, Dreamtech Press
3. C Programming & Data Structures, E. Balagurusamy, TMH

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. II Semester

(25ES2EE104) ELEMENTS OF ELECTRICAL AND ELECTRONICS ENGINEERING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Mathematics, Physics

COURSE OBJECTIVES:
- To recognize different circuit reduction techniques
- To understand the construction of electrical equipment and operation
- To practice the techniques to control and assess electrical machines and installations
- To learn principles of operation, construction and characteristics of various electronic devices

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Apply different network reduction techniques to solve and analyze electrical circuits
CO-2: Realize the compatibility of electrical machines in different engineering fields & Installations
CO-3: Control different electrical machines and evaluate their performance
CO-4: Apply basic electronics device in real time applications

COURSE ARTICULATION MATRIX:
*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	2	2	1	-	-	-	-	1	-	-	-	2	1
CO-2	2	2	1	-	-	-	-	1	-	-	-	2	1
CO-3	2	2	1	-	-	-	-	1	-	-	-	2	1
CO-4	2	2	1	-	-	-	-	1	-	-	-	2	1

LIST OF EXPERIMENTS:

1. Demonstration of safety precautions, measuring instruments, Multimeters, Function Generator, RPS, CRO
2. Demonstration of LT switchgear components
3. Verification of KVL & KCL
4. Verification of the superposition theorem
5. Analysis of series RL, RC, and RLC circuits
6. Control of voltage of an alternator by its excitation
7. Load test on 1- φ transformer

8. Brake test on 3-φ induction motor
9. Speed control of Dc shunt motor
10. Diode Characteristics
11. Transistor CE Characteristics (Input and Output)
12. Half Wave & Full Wave Rectifiers without filter

TEXT BOOKS:

1. Basic Electrical Engineering, D. C. Kulshreshtha, 2nd Edition, TMH, 2019
2. Basic Electrical Engineering, P. Ramana, M. Suryakalavathi, G. T. Chandra Sekhar, 1st Edition, S. Chand, 2018
3. Electronic Devices and Circuits, S. Salivahanan and N. Suresh Kumar, 3rd Edition TMH, 2019

REFERENCES:

1. Electrical and Electronics Technology, E. Hughes, 10th Edition, Pearson, 2010
2. Electrical Engineering Fundamentals, Vincent Deltoro, 2nd Edition, Prentice Hall India, 1989
3. Electrical and Electronics Measurements and Instrumentation, A. K. Sawhney, 3rd Edition, Dhanpat Rai & Co., 1983
4. Basic Electrical Engineering, D. P. Kothari and I. J. Nagrath, 3rd Edition, Tata McGraw-Hill, 2010
5. Engineering Circuit Analysis, William Hayt and Jack E. Kemmerly, 8th Edition, McGraw-Hill, 2013

ONLINE RESOURCE:

1. https://nptel.ac.in/courses/117106108

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. II Semester

(25MN6HS102) ENVIRONMENTAL SCIENCE

TEACHING SCHEME		
L	T/P	C
2	0	0

EVALUATION SCHEME		
SE-I	SE-II	TOTAL
50	50	100

COURSE PRE-REQUISITES: Basic Knowledge on Environmental Issues

COURSE OBJECTIVES:
- To recognize the principles of environment
- To identify the sources and causes of environmental pollution
- To analyze the causes of climate change and identify climate change mitigation and adaptation technologies
- To list out the benefits in creating circular economy and sustainability
- To emphasize the environmental conventions, policies and regulations

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Gain knowledge about principles of Environment and allied problems
CO-2: Identify the causes and suggest remedial measures for Environmental Pollution
CO-3: Appraise the mitigation measures and adaptation technologies related to climate change
CO-4: Familiarize with the importance of circular economy towards sustainability
CO-5: Relate and implement environmental policies and regulations

COURSE ARTICULATION MATRIX:
(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial****)*

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	2	2	1	-	1	2	2	1	1	1	2	1	1
CO-2	2	2	1	-	1	2	2	1	1	1	2	1	1
CO-3	2	2	1	-	1	2	2	1	1	1	2	1	1
CO-4	2	2	1	-	1	2	2	1	1	1	2	-	-
CO-5	-	1	1	-	-	2	1	1	1	1	-	-	-

MODULE 1:
Introduction to Environmental Science: Importance and Scope of Environmental Science. Overview of Environment & its components. Ecosystem function and uses. Biodiversity – Types and Values.

MODULE 2:
Synergy With Environment: Relationship between environmental science & society – Influence of Industry & infrastructure on environment (Specific case studies). Sources, Impacts and prevention of air, water and soil pollution.

MODULE 3:
Climate Change: Science behind climate change – factors responsible for climate change, Role of greenhouse gases – Global temperature rise & its impact on

environment & human health. Carbon footprint – Briefing on Paris agreement, Identify key sectors for low carbon footprint. Climate change mitigation & adaptation strategies.

MODULE 4:

Moving Towards Sustainability: EIA and EMP - importance, Sustainable agriculture – Importance of Organic farming and hydroponics. Role of AI & IOT for efficient management of environmental issues – Health, air, water, and soil. Sustainable living practices – Principles and methods (Linear Economy vs circular economy). Green Buildings.

MODULE 5:

Policies and Regulations: International conventions: Montreal Protocol, COP15, Kyoto protocol (Salient Features). National Environmental Policies – Salient features of EPA act, Air act, Water Act, Forest and Wildlife act. National Green Tribunal Act (2010)- Features and Significance. Extended producer responsibility. Innovative approaches to waste management: Plastic, tyre and batteries.

TEXT BOOKS:

1. Living in the Environment, G. Tyler Miller, Scott E. Spoolman, 20th Edition, Cengage Learning
2. Environmental Studies, Rajagopalan, Oxford University Press, 2005
3. Introduction to Climate Change, Andreas Schmittner, Oregon State University, 2018 https://open.oregonstate.education/climatechange/

REFERENCES:

1. Green Development: Environment and Sustainability in a Developing World, Bill Adams, 4th Edition, Routledge Publishers, 2021
2. Fixing Climate, Robert Kunzig & Wallace S. Broecker, Profile Books Publisher, 2009
3. Plastic Waste and Recycling-Environmental Impact, Societal Issues, Prevention and Solutions, 1st Edition, edited by Trevor Letcher, Academic Press, 2020
4. Waste Management Practices: Municipal, Hazardous, and Industrial, John Pichtel, CRC Press, 2020
5. Environmental Science: A Global Concern, William P. Cunningham, Mary Ann Cunningham, McGraw-Hill Education, 2022

ONLINE RESOURCE:

1. https://www.google.co.in/books/edition/Waste_Management_Practices/pG3SBQAAQBAJ?hl=en&gbpv=1&pg=PA1&printsec=frontcover